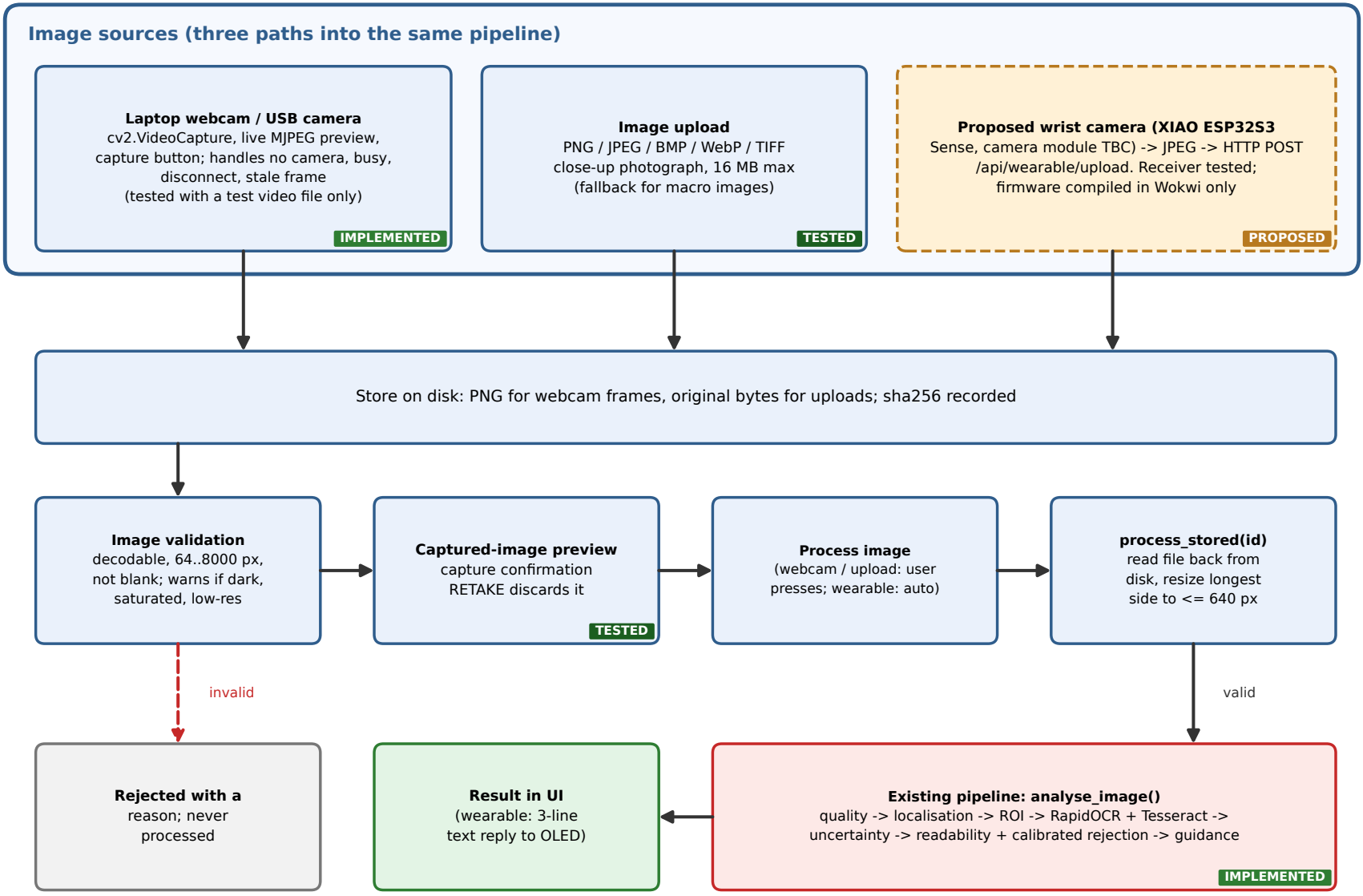Camera sensing and image-acquisition workflow